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- 1.) Java is a high-level programming language developed by James Gosling and his team and released in 1995. It is designed to be simple and platform-independent as it can run on any operating system with a JVM. Another key feature is its object-oriented nature which supports encapsulation, inheritance, polymorphism, and abstraction (core principles of OOP). It also provides automatic memory management through garbage collection, has built-in security features, includes robust exception handling, and provides extensive standard libraries. Java is very versatile because it is used across many industries for mobile applications, enterprise systems, web applications, games, database connections, and more. Aside from its reliability, it is open-source and free as well.
- 2.) It is platform independent because a Java program can run on any operating system by compiling the source code into an intermediate format called "bytecode" using the Java compiler instead of the machine's code. This source code can be executed on any device that has a compatible Java Virtual Machine (JVM) installed. This allows developers to write the code once without needing to recompile for different operating systems.
- 3.) The main components of a Java program are package, class, method, variable, and statement. A package simply groups related classes together to keep a code organized. Class is a building block of a Java program that acts as a blueprint for creating objects and contains the program's data and actions. Method is a block of code that performs a specific task and defines the actions of an object. Variables are used to store data that the program can use and change. It has different data types such as string and int. And lastly, a statement is a single instruction that performs an action. It usually ends with a semicolon and is executed by the program.

4.) JDK (Java Development Kit) is basically used to build Java programs. It includes tools like the compiler (javac) and also contains the JRE. Meanwhile, JRE (Java Runtime Environment) includes the JVM and provides the tools and libraries needed to run Java programs, but does not have development tools like the compiler or debugger. And JVM (Java Virtual Machine) is a virtual machine that actually runs the compiled Java bytecode. It is responsible for interpreting the code and executing it.

5.) In a compiled language, the entire program is converted at once into machine code by a compiler into an executable file before it runs. In this language, the execution process is generally faster. On the other hand, a program is read and executed line-by-line by an interpreter during runtime in an interpreted language. Based on what I've gathered online, it appears that Java is a combination of both. When we write Java code, it is first compiled by the Java compiler (javac) into bytecode to convert the human-readable code into a form that a computer can understand. This step makes Java a compiled language. The JVM then takes the compiled bytecode, interprets it at runtime, and translates it into a machine code that a computer can execute. Modern JVMs often use Just-In-Time (JIT) compilation to dynamically convert bytecode into machine code. This runtime execution step is what makes Java an interpreted language as well.